INCH-POUND

MIL-T-27/212B
20 April 1993
SUPERSEDING
MIL-T-27/212A
8 May 1986

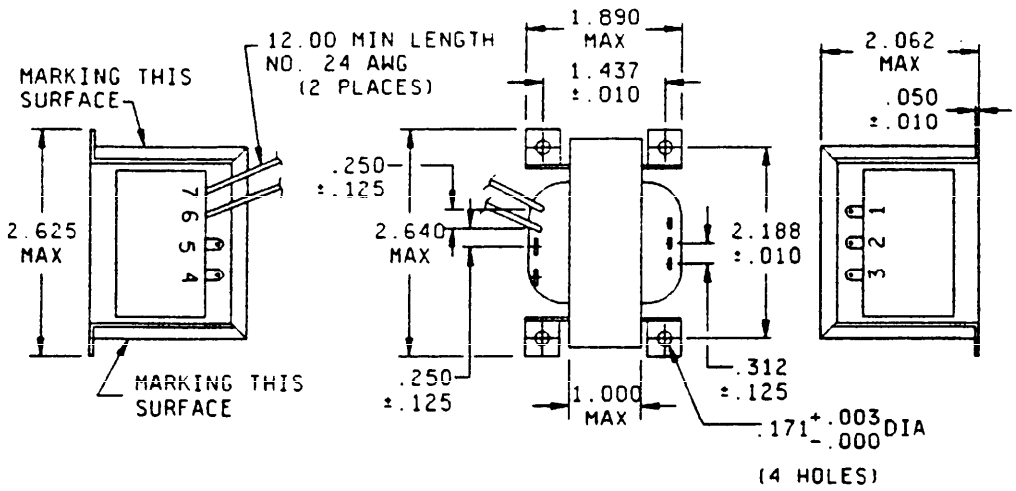
MILITARY SPECIFICATION SHEET

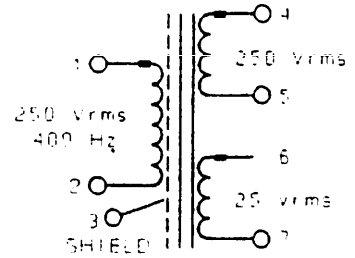
TRANSFORMER, POWER, 50 VA, TF5T032Z

(B) Inactive for new design after the date of this specification.

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-T-27.





WIRING VOLTAGE 250 V(EACH WINDING)
ALTITUDE 10,000 FEET MAX

CIRCUIT DIAGRAM AND MARKING

Inches	mm	Inches	mm
.003	0.08	1.437	36.50
.010	0.25	1.890	48.01
.050	1.27	2.062	52.37
.125	3.18	2.188	55.58
.171	4.34	2.625	66.68
.250	6.35	2.640	67.06
.312	7.52	12.000	304.80
1.000	25.40		

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for information only.
3. Marking shall be on the sides of the case.

FIGURE 1. Dimensions and configurations.

(B) denotes changes

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REQUIREMENTS: (When numbers in parentheses, i.e., (1-2) are used, they indicate the winding and the extreme terminals of the windings.)

Electrical ratings:

Primary (1-2): 50 voltamperes, 250 volts rms, 400 Hz.

Secondary:

(4-5): 250 volts rms, 0.2 ampere rms.
(6-7): 25 volts rms, 0.01 ampere rms.

Working voltage:

(1-2): 250 volts rms.
(4-5), (6-7): 250 volts rms.

Design and construction:

Dimensions and configuration: See figure 1.

Duty cycle: Continuous.

Case: Encapsulated.

Material: Epoxy.

Terminals: (5) Solder lugs (copper, tin-plated), and (2) insulated wire leads (No. 24 AWG, type E in accordance with MIL-W-16878).

Height: Lugs (.375 inch (9.53 mm) maximum); leads (12 inches (304.8 mm) minimum length).

Weight: 1.25 pounds maximum.

Operating temperature range: -55°C to +170°C.

Altitude: 70,000 feet maximum.

Terminal strength: MIL-STD-202, method 211, test condition A, 5 pounds.

Dielectric withstanding voltage (each winding):

At sea level: 1,000 volts rms.
At reduced barometric pressure: 312.5 volts rms.

Electrical characteristics:

No load: With 250 volts rms and 400 Hz in (1-2), and current in (1-2) at 75 milliamperes rms maximum, power in (1-2) shall not exceed 1.0 watt maximum. Also, voltage across (4-5) shall be 250 volts rms ±10 percent and voltage across (6-7) shall be 25 volts rms ±10 percent.

Rated load: With 250 volts rms and 400 Hz (1-2) and rated current in all secondaries:

Voltage across (4-5): 250 volts rms +10 percent, -20 percent.
Voltage across (6-7): 25 volts rms +10 percent, -20 percent.

DC resistance:

(1-2): 6.0 ohms ±20 percent.
(4-5): 7.0 ohms ±20 percent.
(6-7): 7.0 ohms ±20 percent.

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Electrostatic shielding: Voltage ratio 5 to 1 at 400 Hz.

Polarity: Additive, with terminals 2 and 4, and 5 and 6 connected.

Temperature rise: 75°C with 250 volts rms, 400 Hz across (1-2) at an ambient temperature of 95°C, full load on terminals (4-5) and (6-7).

Vibration, high frequency: MIL-STD-202, method 204, test condition D.

Salt spray (corrosion): MIL-STD-202, method 101, test condition B.

(B) Quality assurance provisions:

Qualification inspection: Not applicable for this specification sheet.

Quality conformance inspection: Groups A and B tests of MIL-T-27 shall be applicable.

Part or Identifying Number (PIN): M27/212-01.

CONCLUDING MATERIAL

Custodians:
Army - ER
Navy - EC
Air Force - 85

Review activities:
Army - AR
Navy - OS, SH
Air Force - 17, 99
DLA - ES

User activities:
Army - ME
Navy - AS, MC
Air Force - 19

Preparing activity:
Army - ER

Agent:
DLA - ES

(Project 5950-0815-21)

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